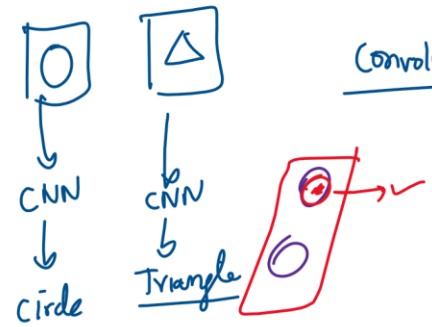
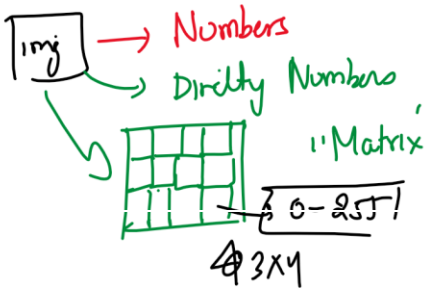
Week 3 DLGenAI



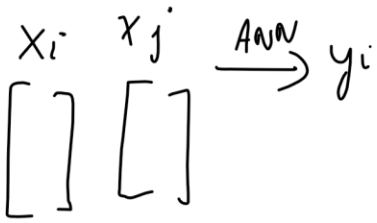
Convolutional Neural Networks

→ Images
- Object Detection
- Classification → Fruits
- Generation
- Medical Diagnosis

Data :- CNN → Images



ANN :- Data

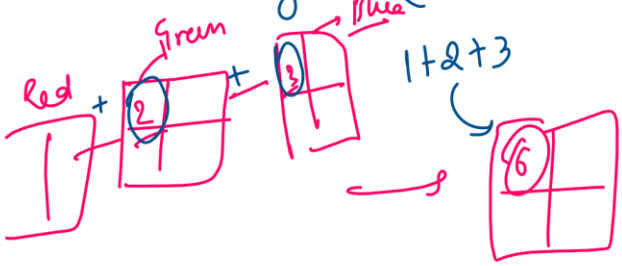


2	141	0
1	3	150

2x3

→ Greyscale Image (Black & White)

→ RGB Image (Coloured Image)



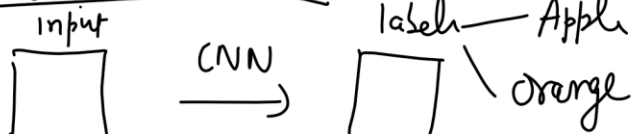
m x n (1 channel)

m x n x 3

layers

Channels

Image Classification



$m \times n$

$= 3 \times 3$

CNN

1. Convolution layer (Kernel) → Identifies features
2. Pooling layer
3. MLP → Classification

Flatten layer

output → Weights → learn

Convolution layer

Kernel → Matrix

Filter

output → Feature Map

input image

3×3

9×1

loss →

Round

Round

"Data Augumentation"

Kernel dimension is decided

Filter

Padding

stride

"stride $\propto \frac{1}{\text{Feature Map Size}}$"

Padding :- Adds a layer of zero on the image

"Padding = 2"

Padding = 1

2×2

Edge Detection (Add more planes]
"Adds no new features"

CNN Formula

$$\text{Output dim} = \frac{\text{Input dim} - \text{Kernel size} + 2\,\text{Padding}}{\text{Stride}} + 1$$

$$\frac{3 - 2 + 0}{1} + 1 = \frac{1}{1} + 1 = 2$$

$$\frac{3 - 2 + 0}{2} + 1 = \frac{1}{2} + 1 = \lfloor 1.5 \rfloor = 1$$

$3 - (1 \times 1)$

$$\begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix}$$

3×2 $\quad 2 \times 2$ $\quad 1 \times 1$ $\quad 1 \times 2$

Stride = 2

P = 0

Length :- Rows $\quad \frac{3 - 2 + 0}{2} + 1 = \frac{1}{2} + 1 = 1.5 = 1$

Cols :- $\quad \frac{2 - 2 + 0}{2} + 1 = 0 + 1 = 1$ "1×1"

mvn

$p=3$ $s=1$ $p=5$ $\begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$ 3×2 $s=1$ 1×1 $\rightarrow$ $\begin{bmatrix} 12 \\ 12 \\ 12 \end{bmatrix}$ 3×2

Row $= \frac{3-1+0}{1} + 1 = \frac{2}{1} + 1 = 3$ $m\times n$

Col $= \frac{2-1+0}{1} + 1 = 1 + 1 = 2$

3 hyp $\rightarrow$ kernel size, stride, Padding

$(2,1,0)$ — dim — 9-5

$(3,1,5)$ — dim: 5-1

① 3×3 $\xrightarrow{\text{Kernel}}$ FM $\begin{bmatrix} 4 & 0 \\ 5 & 1 \end{bmatrix}$ $\xrightarrow{\text{Pooling}}$ FM2 $\begin{bmatrix} 4 & 0 \\ 5 & 1 \end{bmatrix}$

2×2

Max Pool $\rightarrow$ 2×2 $s=1$ $\rightarrow$ $\boxed{5}$ 1×1

Min Pool $\longrightarrow$ $\boxed{0}$

Avg Pool "Mean" $\frac{10}{4} = \boxed{2.5}$

$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 9 & 3 \\ 1 & 2 & 3 & 4 \\ 0 & 0 & 9 & 3 \\ 4 & 0 & 5 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 9 \\ 4 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 2 \\ 1 \\ 4 \\ 9 \end{bmatrix}$

2×2 stride $= 2$ = Pooling layer has no parameters

Why Pool? $\rightarrow$ ① Reduces Parameters

② Extract most relevant features

"True" 55%
pz 71%

Pipeline { I → C → P → MLP

I → C_1 → P_1 → C_2 → P_2 → ... C_n → P_n → MLP

After C1 & P1

$\begin{bmatrix} 2 & 0 \\ 4 & 9 \end{bmatrix}$ → Flattening (Row Wise)

⇒ $\begin{bmatrix} 2 \\ 0 \\ 4 \\ 9 \end{bmatrix}$ → Input to mlp

4x1

$\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

Wxtb 4→1

2 0 w_1
9 0 w_2
4 0 w_3
9 0 w_4

[6]

Fol. of activ

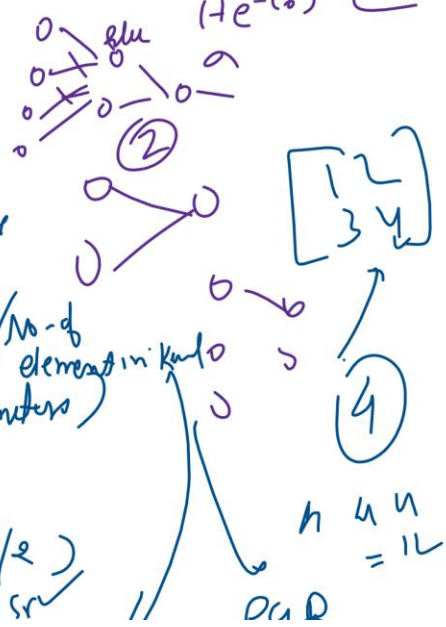
$= [2+0+4+0] = [6]$

$a(6) = \frac{1}{1+e^{-(6)}} = \boxed{0.7}$

3D

MLP → Same

[ ] → [ ]



How to calculate total params

1. CNN → Not hyperparameters

(1) Conv → 2x2 = 4 → Size/No. of elements in kernel

(2) ~~Pool → 0~~ (No parameters)

(3) ~~Flat → 0~~

(4) MLP → Same (Week 1/2)

→ SR

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

4

h 4 4 = 16

DCAD

CNN = 1+2+3+4

= 1+4 MLPt conv 3x3 Grey

M. fel. X3

Grey $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \rightarrow 4 + 1 = 5$

M bias 10 bias

5N 10 N +10

5x10=50 / 60 with bias

RGB $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ (W1, W2, W3) $\rightarrow 3 \times 4 = 12$, $+3 = 15$

"1 bias per Kernel"

Exam → Weights

→ Parameter / Trainable Parameters (W+b)

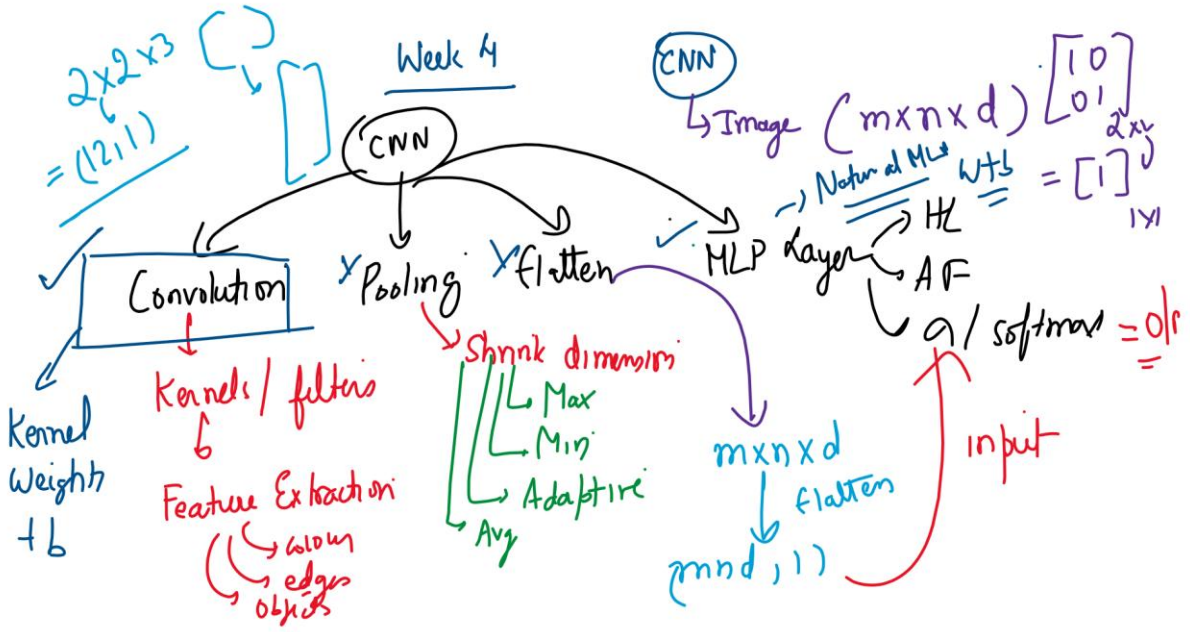
→ Parameters / TP (Ignore bias) = W

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3x3 = 9 Weights

T = W+B = 9+1 = 10

m x n = m m



$$O_{dim} = \frac{I_{dim} - K + 2P}{S} + 1$$